2. **Generalized Tikhonov regularization.** Consider the Abel problem for $x(s) = s^{3/4} - 1$. Discretize K by a matrix $A \in \mathbb{R}^{n\times n}, n = 1024$, and discretize similarly $x(s)$ and $y(t) = Kx(t)$ by $x, y \in \mathbb{R}^n$. Add Gaussian noise e with relative magnitude 10%, i.e. $\|e\|_2 = 0.1\|y\|_2$, to get $y^\delta = y + e$. Denote by $D \in \mathbb{R}^{(n-1)\times n}$ the forward difference matrix approximating $Dx \approx x'(s)$. (Make sure to scale with n.)

(a) Since $x \in H^1(0,1)$ it makes sense to apply H^1 regularization in the inverse problem. We approximate the H^1-norm by $\|x\|_2^2 + \|Dx\|_2^2$, with D being the finite difference first derivative matrix.
Show that the problem

$$x_\alpha = \arg\min_x \overbrace{\|Ax - y^\delta\|_2^2 + \alpha(\|x\|_2^2 + \|Dx\|_2^2)}^{J(x)} \tag{1}$$

has a unique solution for any $\alpha > 0$.

we want to show, that $J(x) = \|Ax-y\|_2^2 + \alpha(\|x\|_2^2 + \|Dx\|_2^2)$ has a unique minimum

1: it holds, that strictly convex functions on $\mathbb{R}^n$ have at most one minimum:

$J(\theta x + (1-\theta)y) < \theta J(x) + (1-\theta)J(y)$
$\Rightarrow$ lets assume x_1, x_2 are different minima for J ($x_1 \neq x_2$)
so $J(x_1) \leq J(x)$ $\forall x \in H$, $i = 1,2$
$\Rightarrow J(\frac{1}{2}(x_1+x_2)) \overset{J\text{ conv}}{<} \frac{1}{2}(J(x_1)+J(x_2)) \overset{\text{since both are minimum}}{=} J(x_i)$ ↯ □

Preparation:

- $\|\cdot\|_2$ convex: we know, that in order to be a norm, 3 conditions must hold:
  1. $\|\cdot\|_2 \geq 0 \;\wedge\; \|x\|_2 = 0 \Leftrightarrow x = 0$
  2. $\|\lambda x\|_2 = |\lambda| \|x\|_2$ $\forall x \in H, \lambda \in \mathbb{K}$
  3. $\|x+y\|_2 \leq \|x\| + \|y\|_2$ $\forall x, y \in H$

  we can leave the absolute value, since $\theta, (1-\theta) > 0$

  $\Rightarrow \|\theta x + (1-\theta)y\|_2 \overset{③}{\leq} \|\theta x\|_2 + \|(1-\theta)y\|_2 \overset{②}{=} \theta\|x\|_2 + (1-\theta)\|y\|_2$ $\forall \theta \in (0,1)$ $\Rightarrow \|\cdot\|_2$ convex

- f convex, g convex, (strictly) mon. increasing $\Rightarrow g \circ f$ convex

  $g(f(\theta x + (1-\theta)y)) \overset{f\text{ conv}, g\uparrow}{\leq} g(\theta f(x) + (1-\theta)f(y)) \overset{g\text{ conv}}{\leq} \theta g(f(x)) + (1-\theta)g(f(y))$

- f, g convex, $\alpha > 0 \Rightarrow f + \alpha g$ conv

  $(f+\alpha g)(\theta x + (1-\theta)y) = f(\theta x + (1-\theta)y) + \alpha g(\theta x + (1-\theta)y)$
  $\overset{f,g\text{ conv}}{\leq} \theta f(x) + (1-\theta)f(y) + \alpha(\theta g(x) + (1-\theta)g(y))$
  $= \theta(f(x) + \alpha g(x)) + (1-\theta)(f(y) + \alpha g(y))$
  $= \theta(f+\alpha g)(x) + (1-\theta)(f+\alpha g)(y)$

- $D \in \mathbb{R}^{n\times m}$ conv.

  $D(\theta x + (1-\theta)y) \overset{D\text{ linear}}{=} \theta Dx + (1-\theta)Dy$

- $f(x) = Ax - y$ conv.

  $f(\theta x_1 + (1-\theta)x_2) = A(\theta x_1 + (1-\theta)x_2) - y\underbrace{(\theta + (1-\theta))}_{=1} \overset{A\text{ linear}}{=} \theta(Ax_1 - y) + (1-\theta)(Ax_2 - y)$
  $= \theta f(x_1) + (1-\theta)f(x_2)$

So we now define our function $J(x) = \|Ax - y^\delta\|_2^2 + \alpha[\|x\|_2^2 + \|Dx\|_2^2]$
with the following helping fct.:

$f_1(x) = \|Ax - y^\delta\|_2$
$f_2(x) = \alpha\|x\|_2^2$
$f_3(x) = \|Dx\|_2$
$g(x) = x^2$
} all convex, f_2 strictly convex

$\Rightarrow J(x) = g(f_1(x)) + \alpha(f_2(x)) + \alpha g(f_3(x))$ which is convex, due to our preparations and because of f_2 even strictly convex

2: we need to show, that there exists a minimum:

since $J(x) \geq \alpha\|x\|_2^2 \overset{\|x\|_2\to\infty}{\longrightarrow} \infty$, we know J is coerciv ($J(x) \overset{\|x\|_2\to\infty}{\longrightarrow} \infty$)
and since J is continuous $\Rightarrow$ there exists a minimum for J